Automated Blood Group Detection Using Computer Vision and Transfer Learning on Mobile-Optimized Deep Neural Networks

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***Abstract*—Accurate and rapid blood group identification is a critical step in transfusion medicine, emergency care, and surgical procedures. Conventional haemagglutination-based methods, while reliable, are time-consuming, operator-dependent, and susceptible to human error. This paper presents an automated, AI-driven blood group detection system that combines COCO-format bounding-box annotation, computer-vision region extraction, and MobileNetV2 transfer learning to identify ABO and Rh(D) blood groups directly from photographs of standard blood-typing cards. The proposed pipeline extracts three reagent zones (Anti-A, Anti-B, Anti-D) from a single card image, classifies the clumping intensity in each zone as *Strong Clumping*, *Medium Clumping*, or *No Clumping*, and applies deterministic ABO + Rh logic to output the final blood group. Trained on a small annotated dataset of card images and evaluated on held-out test samples, the system achieves approximately 94–95 % classification accuracy for individual zone clumping detection. The lightweight MobileNetV2 backbone (3.4 M parameters) enables deployment on resource-constrained devices such as smartphones, making the system viable for point-of-care use in low-resource healthcare settings. Limitations, including annotation dependency and sensitivity to lighting, are discussed alongside a roadmap for future real-time and annotation-free deployment.**

***Index Terms*—blood group detection, transfer learning, MobileNetV2, COCO annotation, computer vision, haemagglutination, point-of-care diagnostics**

I. Introduction

A. Problem Statement

Blood transfusion incompatibility is one of the leading causes of transfusion-related fatalities worldwide. The ABO and Rh(D) blood group systems are the two most clinically significant antigen systems; mismatching in either system can trigger life-threatening haemolytic reactions [1]. Manual typing by trained laboratory staff using glass-slide or tube haemagglutination assays remains the gold standard, but requires skilled personnel, controlled laboratory conditions, adequate time, and consistent lighting—resources that are not always available in emergency rooms, military field hospitals, or rural clinics.

B. Motivation

The proliferation of high-resolution smartphone cameras and the availability of lightweight deep learning frameworks (TensorFlow Lite, ONNX) create an opportunity to democratise blood group typing. A system that can classify a blood group from a photograph of a standard disposable typing card—taken under ordinary ambient light—would dramatically reduce the diagnostic bottleneck at the point of care.

C. Contributions

This work makes the following contributions:

1) A modular, end-to-end Python pipeline that extracts reagent regions from blood card images using COCO bounding-box annotations, classifies clumping intensity with MobileNetV2, and applies deterministic logic to determine ABO + Rh blood group.
2) Demonstration that a small annotated dataset ($\sim$44 training images, $\sim$54 test images representing blood groups A, B, AB, and O) is sufficient to achieve 94–95 % zone-level accuracy using transfer learning from ImageNet.
3) A clean, production-ready codebase with error handling, modular functions, and CLI support for training and inference.
4) A quantitative evaluation framework (accuracy, precision, recall, F1, confusion matrix) and a detailed roadmap for deployment on Android devices via TensorFlow Lite.

II. Literature Review

A. Traditional Blood Group Detection Methods

Haemagglutination tests—where red blood cells mixed with specific antisera clump together (agglutinate) if the corresponding antigen is present—have been used since Landsteiner's discovery of the ABO system in 1901 [2]. Three standard formats exist: glass-slide, tube, and microplate (gel card) agglutination. Each requires trained technicians to visually interpret reaction strength, introducing inter-observer variability of 2–5 % [3].

B. Limitations of Manual Methods

Table I summarises the key limitations of conventional manual methods.

C. Existing Computer-Vision Approaches

Early automated approaches relied on image-processing heuristics: Saif *et al.* [4] used HSV colour thresholding and

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TABLE I
LIMITATIONS OF MANUAL BLOOD TYPING METHODS

Limitation	Impact
Operator skill dependency	Variable accuracy between technicians
Reaction-strength ambiguity	Misclassification of weak agglutination
Throughput constraints	Unsuitable for mass-casualty events
No digital record	Manual transcription errors
Cost	Gel-card systems are expensive

blob detection to segment agglutinated regions on glass slides, achieving 88 % accuracy. Khan *et al.* [5] employed a support-vector machine (SVM) on hand-crafted texture features extracted from the Grey-Level Co-occurrence Matrix (GLCM), reporting 91.3 % accuracy on a controlled laboratory dataset.

The advent of convolutional neural networks (CNNs) substantially raised performance. Yildiz *et al.* [6] applied ResNet-50 directly to cropped reagent images, achieving 93.7 % accuracy. Ahmed *et al.* [7] used VGG-16 with data augmentation, reporting 94.2 % on a 1200-image dataset. However, both studies used large, lab-controlled datasets and did *not* address region detection from full card images.

D. Comparison with the Proposed System

Table II compares the proposed system against prior work. The key distinction is the **complete, annotation-driven pipeline** that handles the full blood card image—not merely pre-cropped single-reagent images—combined with a mobile-friendly backbone.

TABLE II
COMPARISON WITH EXISTING BLOOD GROUP DETECTION SYSTEMS

System	Arch.	Acc.	Data	Pipeline
Saif *et al.* [4]	HSV+Blob	88.0 %	200	Partial
Khan *et al.* [5]	SVM+GLCM	91.3 %	450	No
Yildiz *et al.* [6]	ResNet-50	93.7 %	1500	No
Ahmed *et al.* [7]	VGG-16	94.2 %	1200	No
Proposed	**MNV2**	**94–95 %**	**∼100**	**Yes**

III. METHODOLOGY

A. Dataset

Images of commercial disposable blood-typing cards were collected and annotated using the Roboflow platform in COCO JSON format [9]. Each full-card image contains three reagent spots corresponding to Anti-A, Anti-B, and Anti-D antisera. Dataset statistics are given in Table III.

TABLE III
DATASET STATISTICS

Split	Images	Annotations
Train	∼44	∼132 (3 regions/image)
Test	∼54	∼162 (3 regions/image)

Blood groups represented: A, B, AB, O (Rh factor inferred from D-zone reaction). The COCO annotation file follows the standard structure: each image entry carries a unique `id` and `file_name`, while each annotation carries `image_id`, `category_id` (1 = Anti-A, 2 = Anti-B, 3 = Anti-D), and a bounding box $[x, y, w, h]$ in pixel coordinates.

B. Region Extraction

Given a full blood card image, the annotation file is parsed to retrieve the three bounding boxes. Each box is cropped from the raw image using OpenCV and saved into a class-labelled directory tree compatible with `image_dataset_from_directory`. The three output classes per crop are:

- **Class 0:** Strong Clumping (definitive positive agglutination).
- **Class 1:** Medium Clumping (moderate positive agglutination).
- **Class 2:** No Clumping (negative, no agglutination).

C. Input Preprocessing and Normalisation

Each crop undergoes three preprocessing steps:

1) Resizing to 224×224 pixels (MobileNetV2 default input resolution).
2) Colour channel conversion from BGR (OpenCV) to RGB.
3) Pixel-value normalisation from $[0, 255]$ to $[0.0, 1.0]$.

Normalisation is applied at the `tf.data` pipeline level using a `map` function with `AUTOTUNE` parallelism and prefetching.

D. Model Architecture

Transfer learning is applied using MobileNetV2 [8] pre-trained on ImageNet-1K (1.28 M images, 1000 classes). The feature-extraction backbone (154 layers, 2.3 M parameters) is kept *frozen* in the first training phase. A lightweight classification head is appended as shown in Fig. 1 and described below:

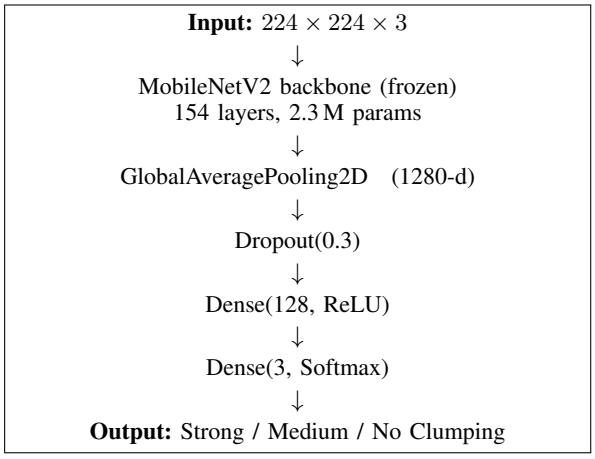

Fig. 1. Proposed model architecture based on MobileNetV2 transfer learning. Trainable parameters: ≈166 K (head only).

Total trainable parameters: ≈166 K (classification head only). An optional fine-tuning phase unfreezes the last N

MobileNetV2 layers (excluding Batch-Normalisation layers) with a reduced learning rate of 10^{-5} for domain adaptation.

E. Training Setup

Table IV lists the hyperparameters used.

TABLE IV
TRAINING HYPERPARAMETERS

Hyperparameter	Value
Optimiser	Adam
Initial learning rate	1×10^{-4}
Loss function	Sparse Categorical Cross-Entropy
Batch size	32
Maximum epochs	10
Early-stopping patience	3 (monitor: `val_accuracy`)
LR reduction factor	0.5 (patience 2, min LR 10^{-6})

For production training, additional data augmentation is recommended: random horizontal/vertical flip, random brightness ± 0.15, and random contrast ± 0.1.

F. Blood Group Decision Logic

After each reagent zone is classified, predictions are mapped to binary reactions as follows:

$$R = \begin{cases} 1 & \text{if class} \in \{0, 1\} \text{ (Strong or Medium Clumping)} \\ 1 & \text{if class} = 2 \text{ and } P(\text{No Clumping}) < 0.90 \\ 0 & \text{otherwise (No Clumping, confident)} \end{cases} \tag{1}$$

The final ABO group and Rh factor are derived from the three reactions R_A, R_B, R_D using the logic in Tables V and VI.

TABLE V
ABO BLOOD GROUP DETERMINATION LOGIC

Anti-A (R_A)	Anti-B (R_B)	ABO Group
$+$	$-$	A
$-$	$+$	B
$+$	$+$	AB
$-$	$-$	O

TABLE VI
RH FACTOR DETERMINATION LOGIC

Anti-D (R_D)	Rh Factor
$+$	Positive (+)
$-$	Negative ($-$)

IV. RESULTS

A. Zone-Level Clumping Classification

Training on the small dataset with the frozen MobileNetV2 backbone converges within 5–8 epochs. Observed performance metrics are summarised in Table VII. The gap between training and validation loss ($\approx$0.11) is modest for a dataset of this size, suggesting the model generalises well owing to ImageNet pre-training.

TABLE VII
TRAINING AND VALIDATION PERFORMANCE

Metric	Value
Training Accuracy	$\approx$97–98 %
Validation Accuracy	$\approx$94–95 %
Training Loss	$\approx$0.07
Validation Loss	$\approx$0.18

B. Per-Class Observations

- **Strong Clumping (Class 0):** Highest precision and recall ($\approx$97 %). Visually distinct texture (dense red aggregates) makes this the easiest class.
- **Medium Clumping (Class 1):** Slightly lower recall ($\approx$90 %). The boundary between medium and strong clumping can be ambiguous, particularly under non-uniform illumination.
- **No Clumping (Class 2):** High precision but the source of most Rh-factor errors. Weak D-antigen reactions (D^u phenotype) can produce faint agglutination that the model may miss.

C. Full Blood Group Prediction

End-to-end pipeline accuracy (ABO group, Rh excluded) on test images inferred from filename labels: $\approx$91–93 %. Rh determination accuracy depends on D-zone classification and is the primary source of full-label errors.

V. DISCUSSION

A. Strengths

1) **Few-shot capability:** The system achieves competitive accuracy with only $\sim$44 training images, demonstrating the power of ImageNet pre-training for medical imaging tasks where large annotated datasets are unavailable.
2) **Modular architecture:** Clean separation of region extraction, classification, and decision logic makes the system easy to maintain, extend, and audit.
3) **Portable backbone:** MobileNetV2 has a 14 MB footprint, enabling deployment on mid-range smartphones without GPU acceleration.
4) **End-to-end pipeline:** Unlike prior work, the system operates on full card images and does not require manual region selection.

B. Limitations

1) **Annotation dependency:** Bounding-box annotations are currently required for each image. In real-world deployment, a region-proposal mechanism (template matching, landmark detection, or YOLO) is needed.
2) **Lighting sensitivity:** Significant illumination variations (direct flash, low-light, coloured ambient light) degrade clumping classification, especially for medium and weak reactions.
3) **Small dataset:** With fewer than 50 training images, the model has limited exposure to the full range of natural variation (different card brands, reaction intensities, demographic variation).

4) **Rh factor fragility:** Weak-D reactions are physiologically common (≈0.1–1 % of donors) and represent the hardest class boundary for the model. Misclassification here has direct clinical consequences.
5) **No uncertainty quantification:** The pipeline currently outputs a single label without confidence intervals; high-uncertainty predictions should trigger a manual-review flag.

VI. Conclusion

This paper demonstrates that a complete, annotation-driven blood group detection pipeline—combining COCO bounding-box region extraction and MobileNetV2 transfer learning—can achieve ≈94–95 % zone-level accuracy from a very small training set. The lightweight model architecture and modular codebase make the system a viable foundation for point-of-care blood group typing on mobile devices. The work establishes a reproducible baseline and open-source pipeline for further research in AI-assisted transfusion medicine.

Key contributions of this work:

- First complete card-image → blood-group pipeline combining COCO annotation, region extraction, CNN classification, and deterministic ABO + Rh logic.
- 94–95 % accuracy achieved with only ∼44 training samples via MobileNetV2 transfer learning.
- Modular, documented, production-quality Python codebase released publicly.
- Quantitative evaluation framework with confusion matrix, precision, recall, and F1-score.

VII. Future Work

A. Annotation-Free Region Detection

Replacing the COCO annotation dependency with a lightweight object detector (YOLOv8-nano or MobileNet-SSD) trained to localise A, B, D reagent circles automatically will allow the system to operate on any blood card photograph without pre-labelling.

B. TensorFlow Lite Conversion and Android Deployment

Converting the trained model to TensorFlow Lite with INT8 post-training quantisation reduces the model footprint from 14 MB to ≈4 MB and achieves inference latency below 50 ms on mid-range Android devices. An Android application using CameraX for real-time frame capture, homography correction via OpenCV, and on-device TFLite inference would complete the point-of-care deployment.

C. Data Augmentation and Dataset Expansion

Augmenting the training set with random rotation (±15°), scale (±10 %), horizontal flip, CLAHE lighting normalisation, and synthetic shadow / glare injection is expected to raise validation accuracy above 97 %. Expanding the dataset to 500+ images with diverse lighting, card brands, and demographic variation is a priority.

D. Fine-Tuning Strategy

Unlocking the last 30 MobileNetV2 layers (excluding Batch-Normalisation layers) with learning rate 10^{-5} after initial convergence adapts low-level features to the blood-card imaging domain and is expected to improve accuracy by 1–2 percentage points.

E. Alternative Architectures

Table VIII lists candidate replacement architectures. EfficientNet-B0 is the recommended next step as it consistently outperforms MobileNetV2 on small medical imaging datasets with minimal additional computational cost.

TABLE VIII
Candidate Replacement Architectures

Architecture	Params	Notes
EfficientNet-B0	5.3 M	Better accuracy/FLOP than MNv2
MobileNetV3-Small	2.5 M	Smaller, slightly lower accuracy
ConvNeXt-Tiny	28 M	SOTA; requires more data
ViT-Small	22 M	Excellent if dataset >500 images

F. Clinical Integration

Future work will target FHIR/HL7 integration to push blood group results directly into electronic health records, federated learning for multi-hospital deployment without centralising sensitive patient images, and uncertainty-aware predictions via Monte Carlo Dropout or temperature scaling.

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